LIST OF EXPERIMENTS

COURSE CODE : CSA13

COURSE NAME : THEORY OF COMPUTATION

1. Write a C program to simulate a Deterministic Finite Automata (DFA) for the given language representing strings that start with a and end with a

main.c

Run ▶

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int n;
    printf("Enter a string: ");
    scanf("%s", s);
    n = strlen(s);
    if (s[0] == 'a' && s[n - 1] == 'a')
        printf("String Accepted");
    else
        printf("String Rejected");
    return 0;
}
```

Output

```
Enter a string: aba
String Accepted
```

main.c

Run ▶

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int n;
    printf("Enter a string: ");
    scanf("%s", s);
    n = strlen(s);
    if (s[0] == 'a' && s[n - 1] == 'a')
        printf("String Accepted");
    else
        printf("String Rejected");
    return 0;
}
```

Output

```
Enter a string: abbaba
String Accepted
```

main.c

Run

Output

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int n;
    printf("Enter a string: ");
    scanf("%s", s);
    n = strlen(s);
    if (s[0] == 'a' && s[n - 1] == 'a')
        printf("String Accepted");
    else
        printf("String Rejected");
    return 0;
}
```

```
Enter a string: abbab
String Rejected
```

2. Write a C program to simulate a Deterministic Finite Automata (DFA) for the given language representing strings that start with 0 and end with 1

main.c

Run

Output

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int n;
    printf("Enter a string: ");
    scanf("%s", s);
    n = strlen(s);
    if (s[0] == '0' && s[n - 1] == '1')
        printf("String Accepted");
    else
        printf("String Rejected");
    return 0;
}
```

```
Enter a string: 0100101
String Accepted
```

3. Write a C program to simulate a Non-Deterministic Finite Automata (NFA) for the given language representing strings that start with b and end with a

main.c

Run

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int n;
    printf("Enter a string: ");
    scanf("%s", s);
    n = strlen(s);
    if (s[0] == 'b' && s[n - 1] == 'a')
        printf("String Accepted");
    else
        printf("String Rejected");
    return 0;
}
```

Output

```
Enter a string: babbbaaa
String Accepted
```

4. Write a C program to simulate a Non-Deterministic Finite Automata (NFA) for the given language representing strings that start with o and end with 1

main.c

Run

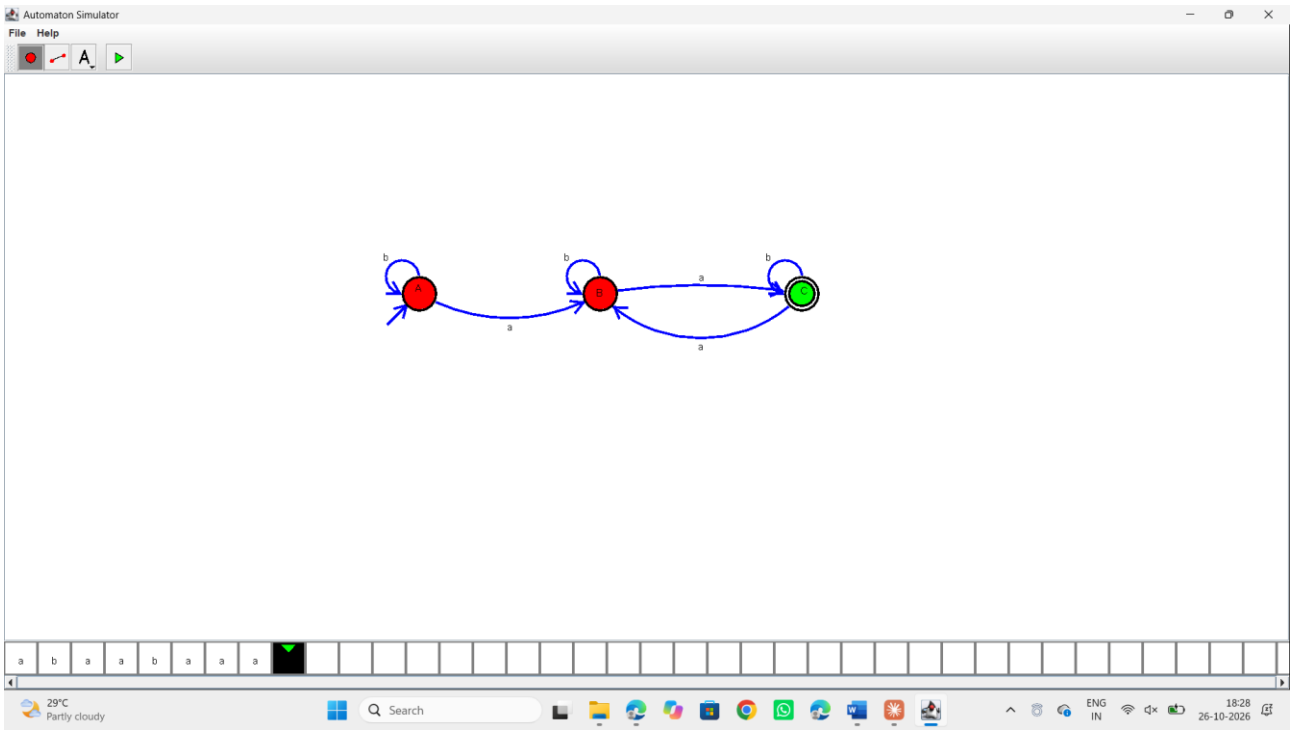
```c
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int n;
    printf("Enter a string: ");
    scanf("%s", s);
    n = strlen(s);
    if (s[0] == '0' && s[n - 1] == '1')
        printf("String Accepted");
    else
        printf("String Rejected");
    return 0;
}
```

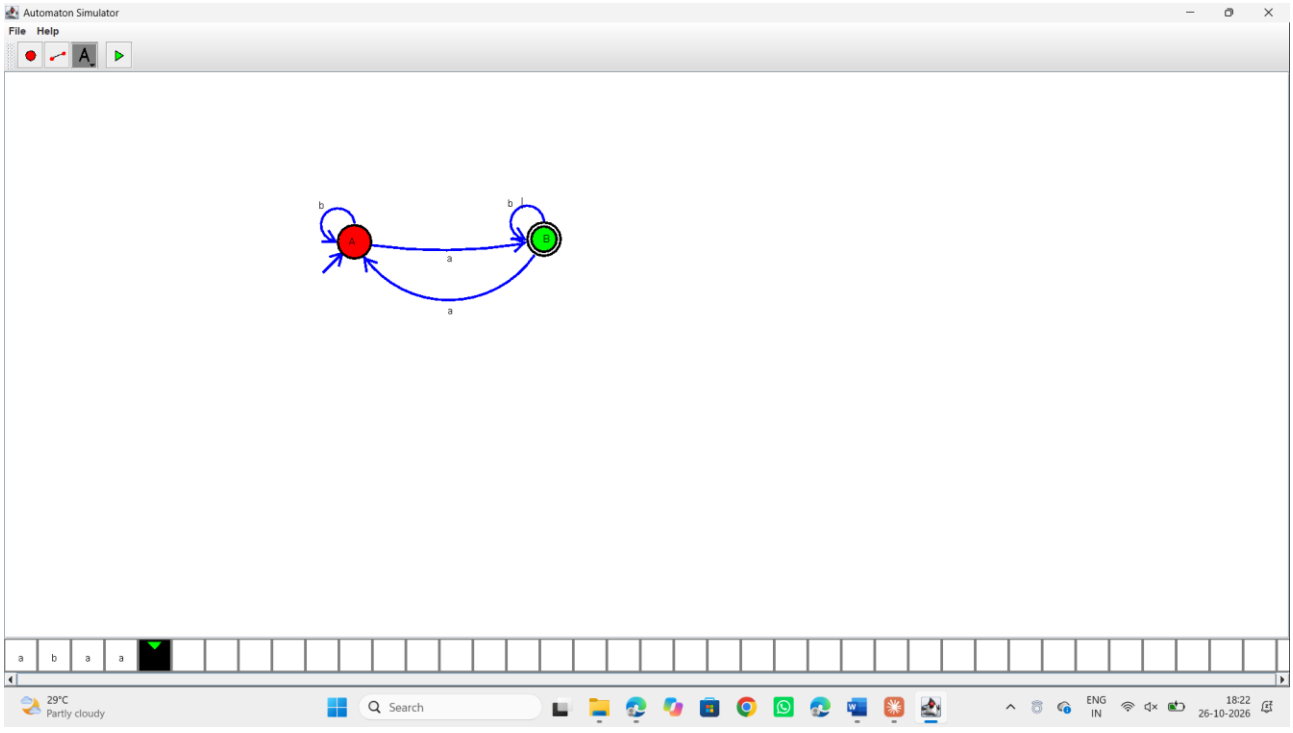
Output

```
Enter a string: 0101
String Accepted
```

5. Design Deterministic Finite Automata using simulator to accept even number of a’s.

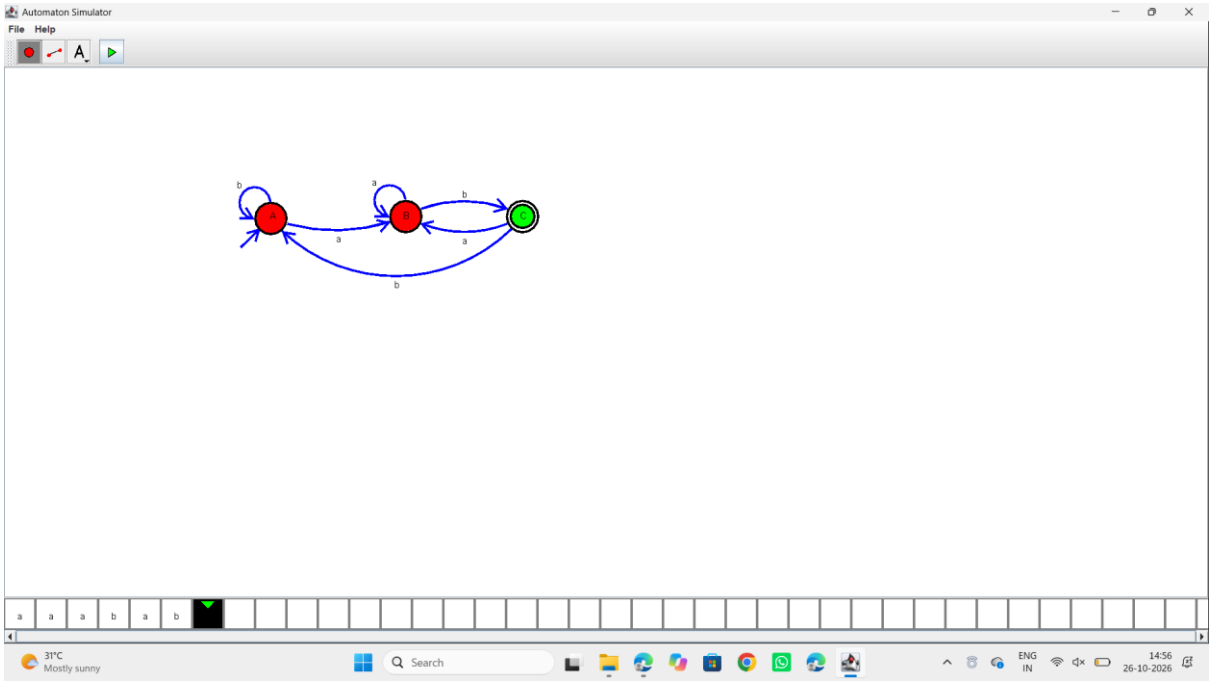


6. Design Deterministic Finite Automata using simulator to accept odd number of a’s

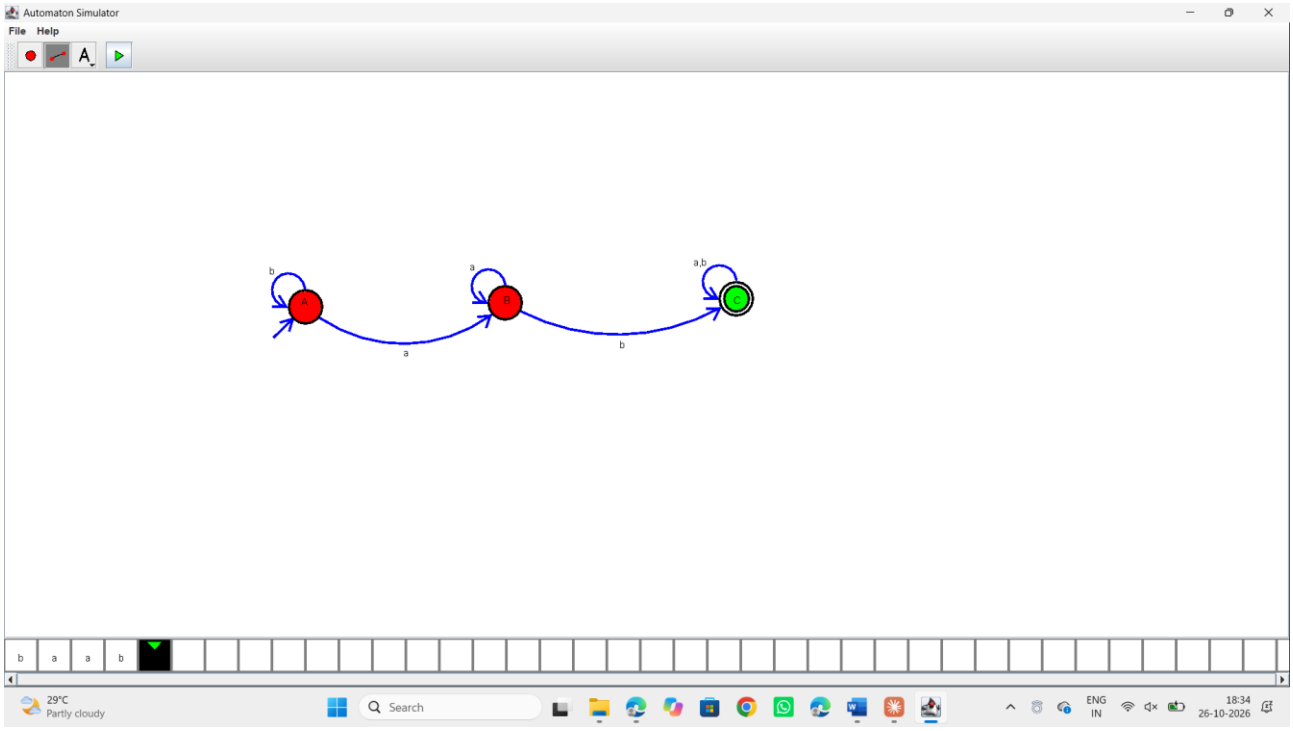


7. Design Deterministic Finite Automata using simulator to accept the string the end with ab over set {a,b)

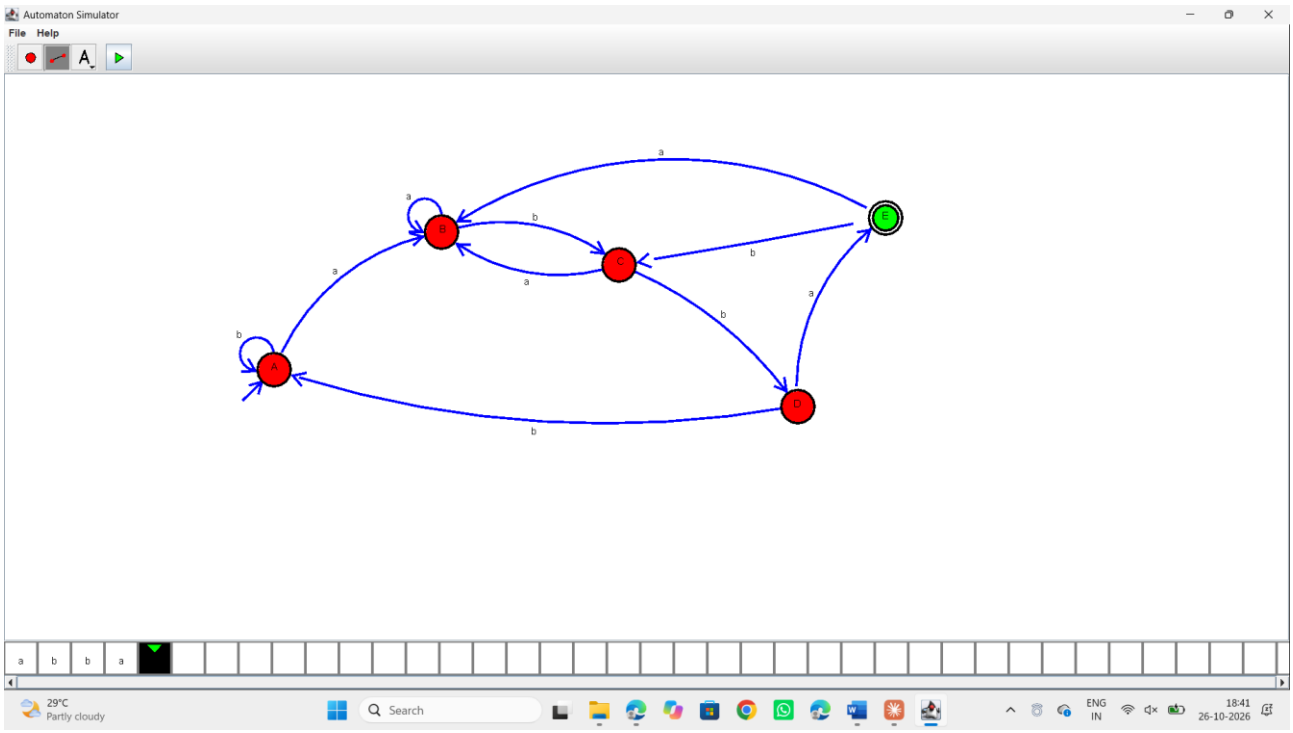
W= aaabab



8. Design Deterministic Finite Automata using simulator to accept the string having ‘ab’ as substring over the set {a,b}



9. Draw a Deterministic Finite Automata for the language accepting strings ending with ‘abba’ over input alphabets ∑ = {a, b}



10. Write a C program to find ε -closure for all the states in a Non-Deterministic Finite Automata (NFA) with ε -moves.

main.c

Run

```c
#include <stdio.h>
int main() {
    int n, i, j, e[10][10];
    printf("Enter number of states: ");
    scanf("%d", &n);
    printf("Enter epsilon transition matrix:\n");
    for (i = 0; i < n; i++)
        for (j = 0; j < n; j++)
            scanf("%d", &e[i][j]);
    for (i = 0; i < n; i++) {
        printf("Epsilon closure of q%d = { q%d", i, i);
        for (j = 0; j < n; j++) {
            if (e[i][j] == 1 && i != j)
                printf(", q%d", j);
        }
        printf(" }\n");
    }
    return 0;
}

```

Output

```
Enter number of states: 3
Enter epsilon transition matrix:
0 1 0
0 0 1
0 0 0
Epsilon closure of q0 = { q0, q1 }
Epsilon closure of q1 = { q1, q2 }
Epsilon closure of q2 = { q2 }
```

11. Write a C program to find ε -closure for all the states in a Non-Deterministic Finite Automata (NFA) with ε -moves.

main.c

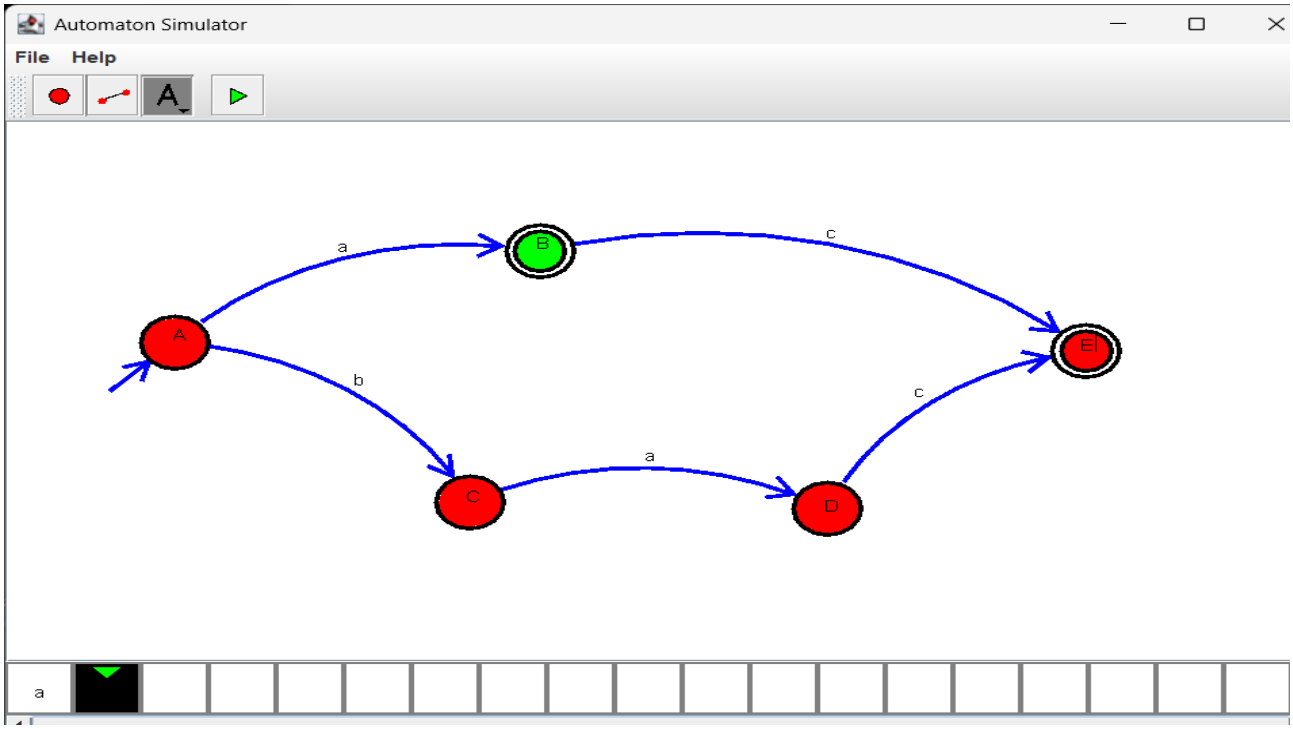
Run ▶

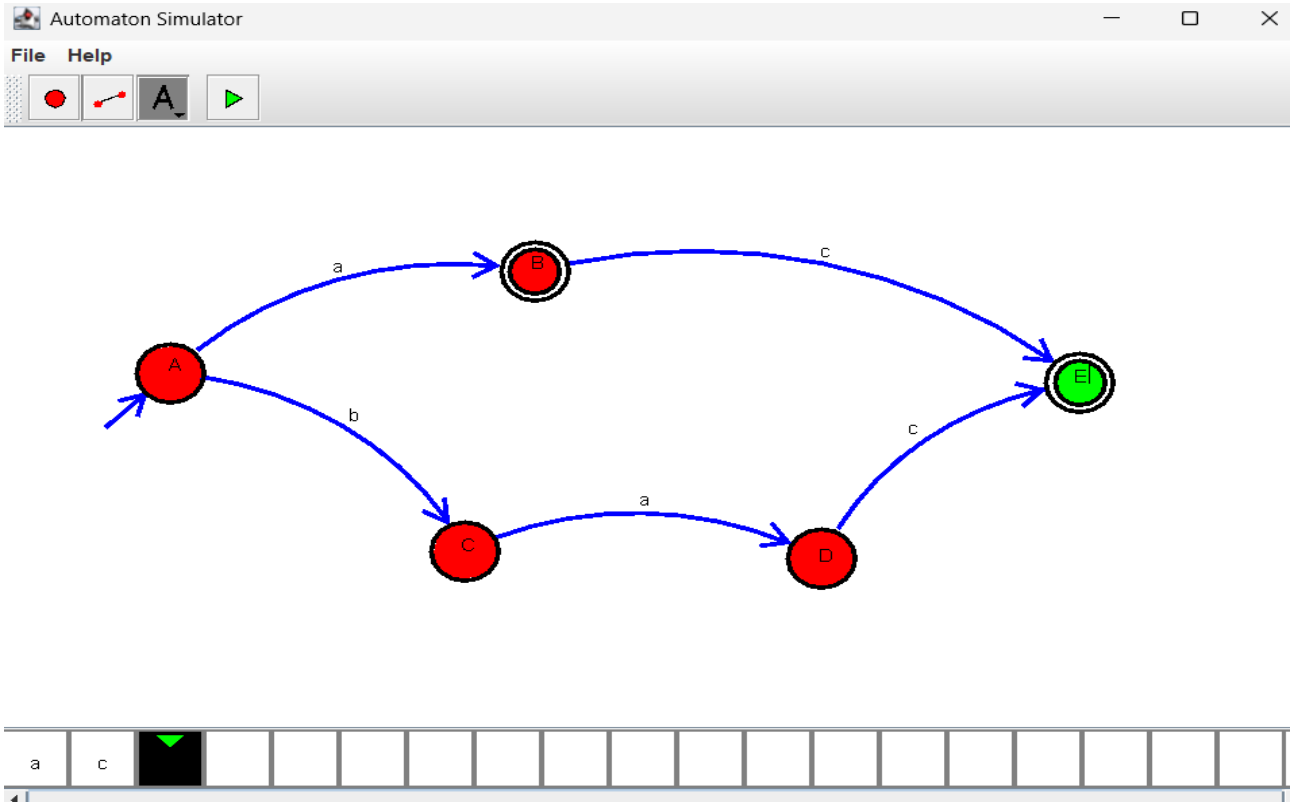
```c
#include <stdio.h>
int main() {
    int n, i, j, e[10][10];
    printf("Enter number of states: ");
    scanf("%d", &n);
    printf("Enter epsilon transition matrix:\n");
    for (i = 0; i < n; i++)
        for (j = 0; j < n; j++)
            scanf("%d", &e[i][j]);
    for (i = 0; i < n; i++) {
        printf("Epsilon closure of q%d = { q%d", i, i);
        for (j = 0; j < n; j++) {
            if (e[i][j] == 1 && i != j)
                printf(", q%d", j);
        }
        printf(" }\n");
    }
    return 0;
}
```

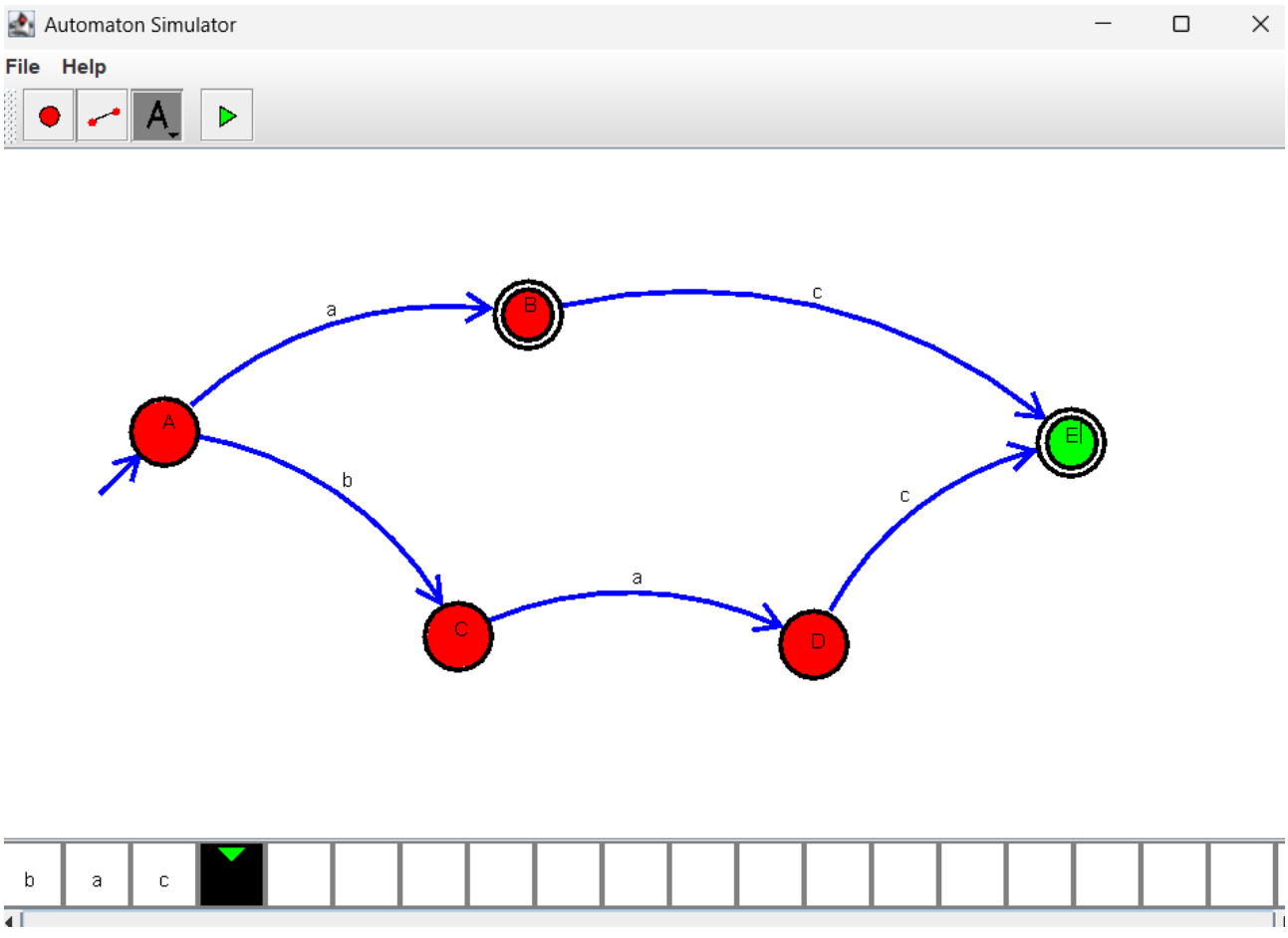
Output

```
Enter number of states: 3
Enter epsilon transition matrix:
0 1 0
0 0 1
0 0 0
Epsilon closure of q0 = { q0, q1 }
Epsilon closure of q1 = { q1, q2 }
Epsilon closure of q2 = { q2 }
```

12. Design Deterministic Finite Automata using simulator to accept the input string “a” ,”ac”,and ”bac”.

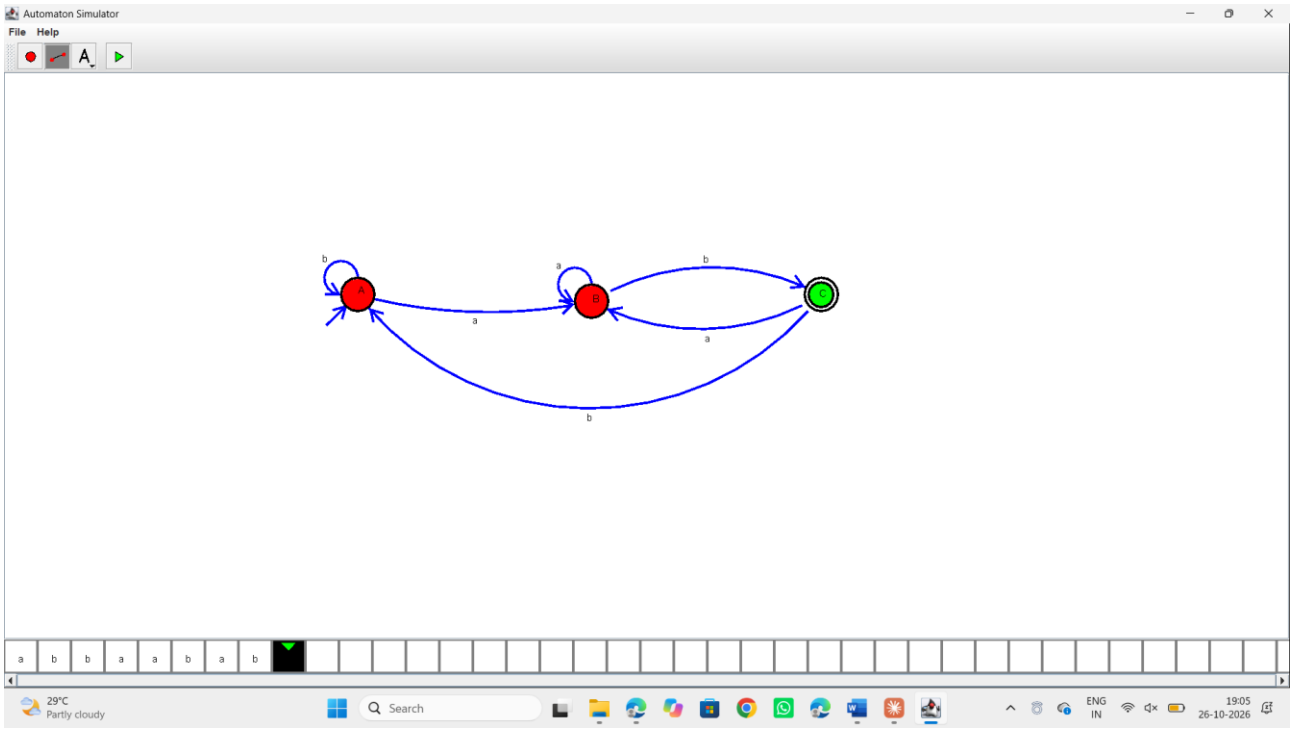



Automaton Simulator
File Help
A
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D
El
a
b
c
a
c
a
c

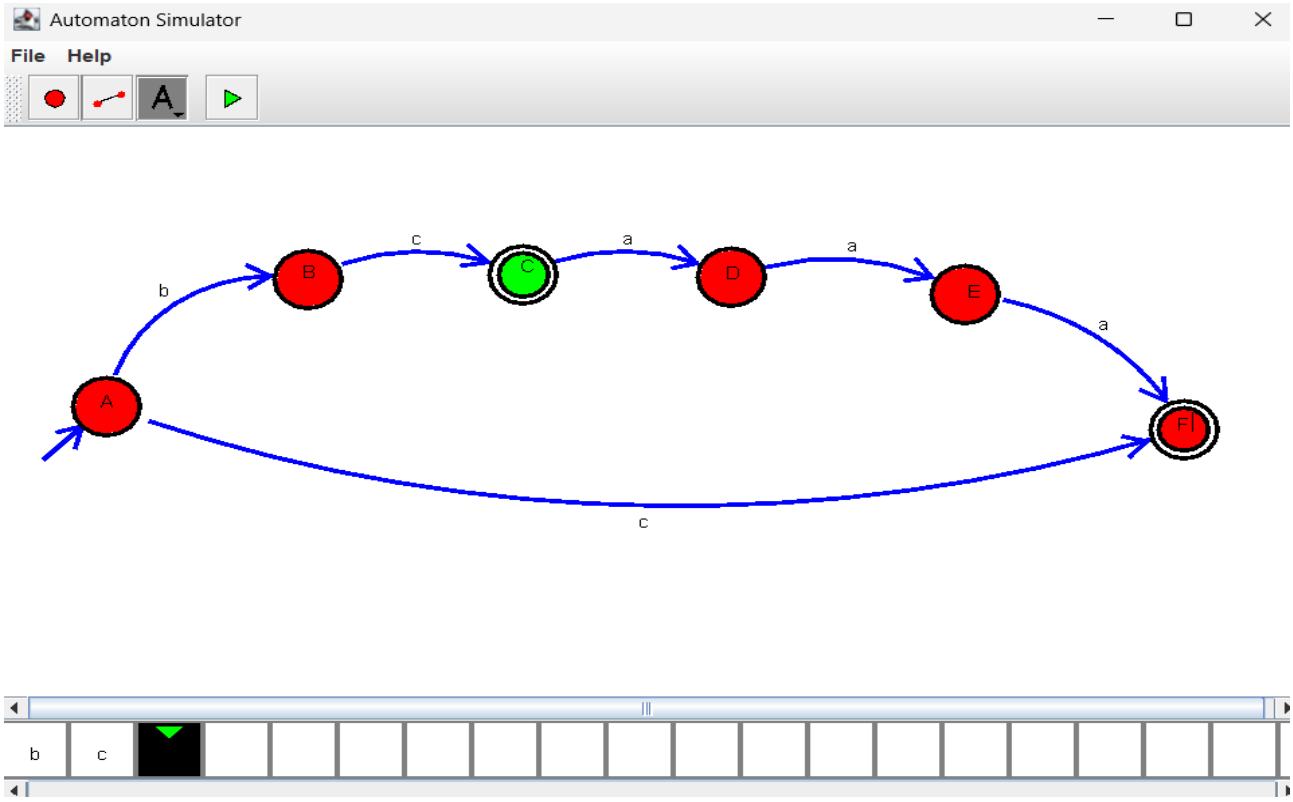

Automaton Simulator
File Help
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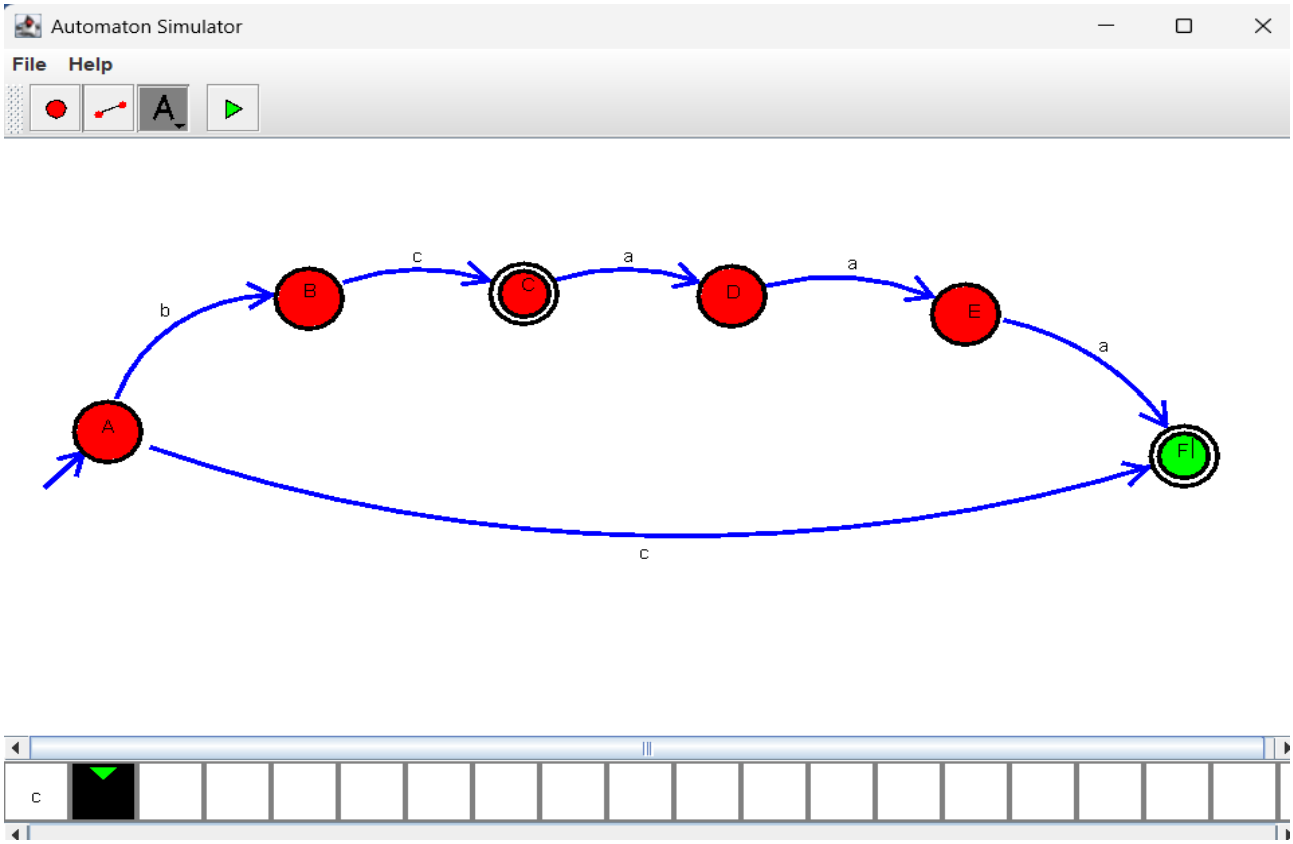
13. Design DFA using simulator to accept the string the end with ab over set {a,b)

W= abbaabab

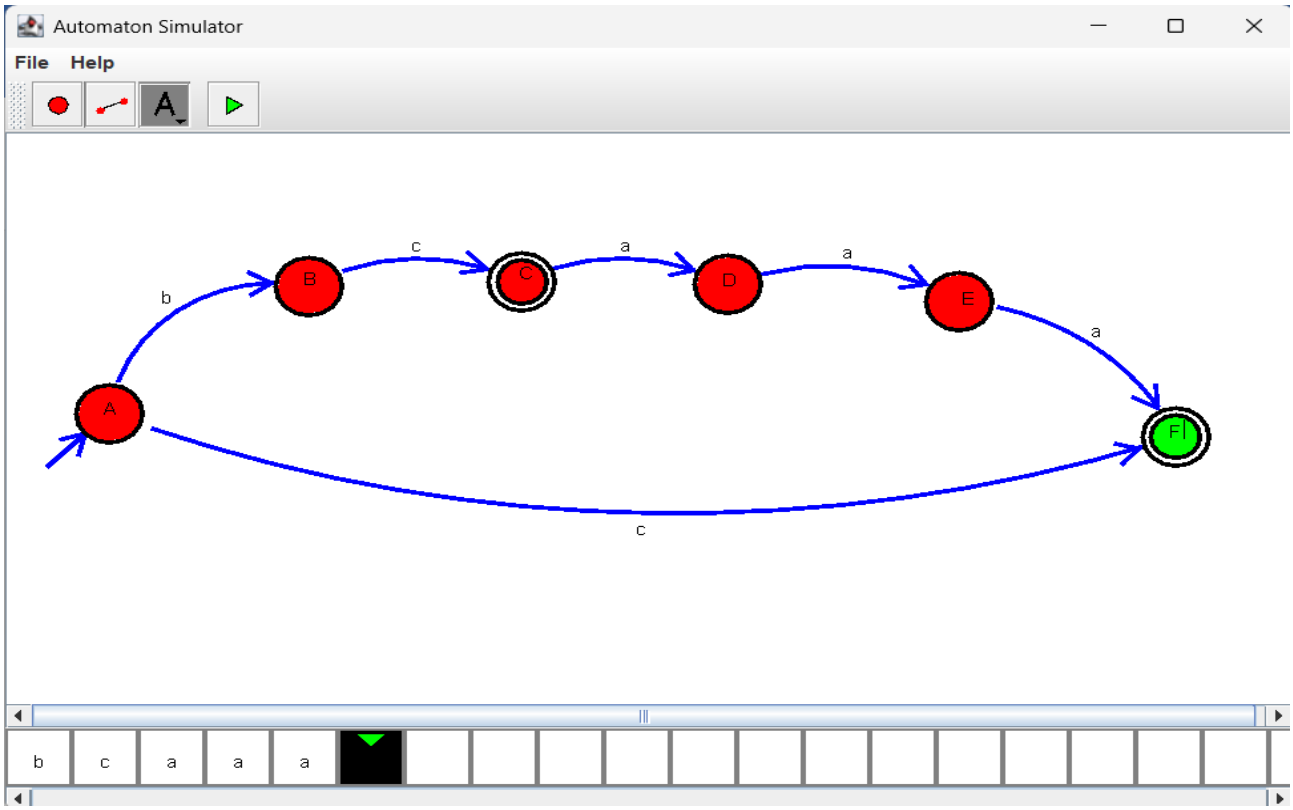


14. Design Deterministic Finite Automata using simulator to accept the input string “bc” ,”c”,and ”bcaaa”.



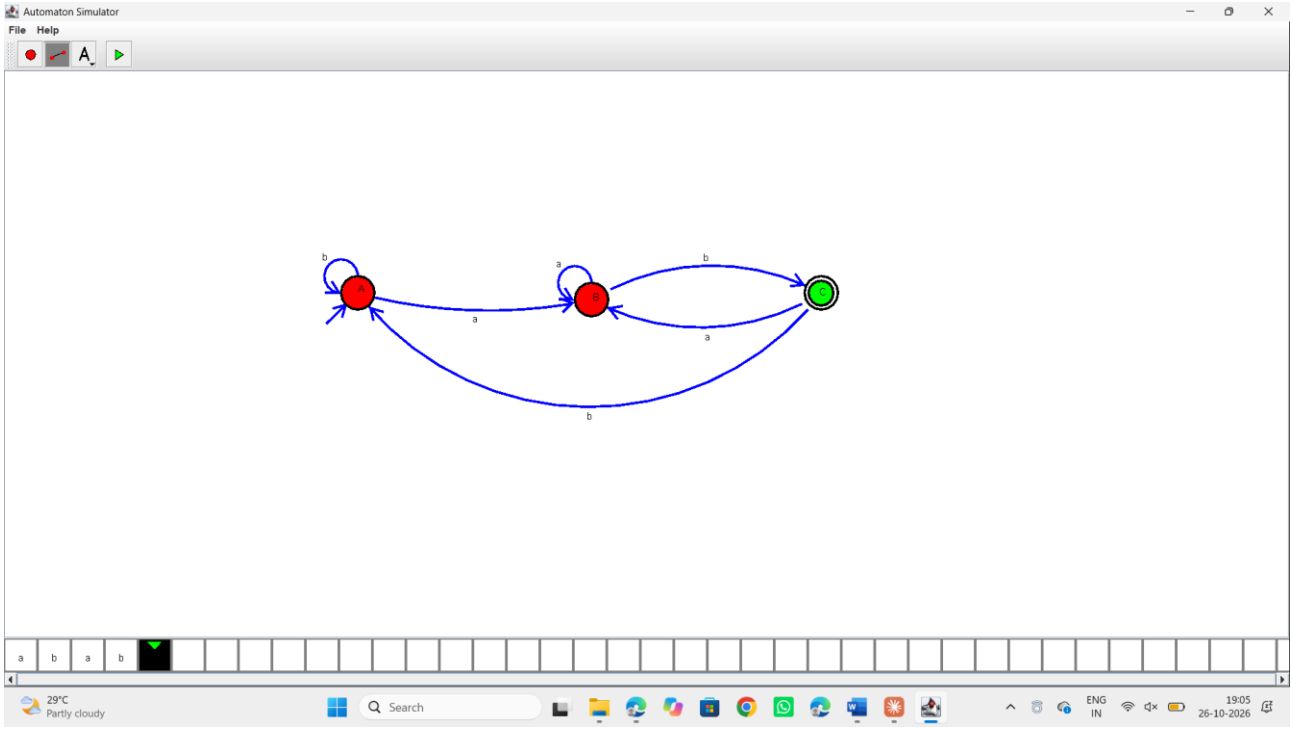

Automaton Simulator
File Help
A
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c

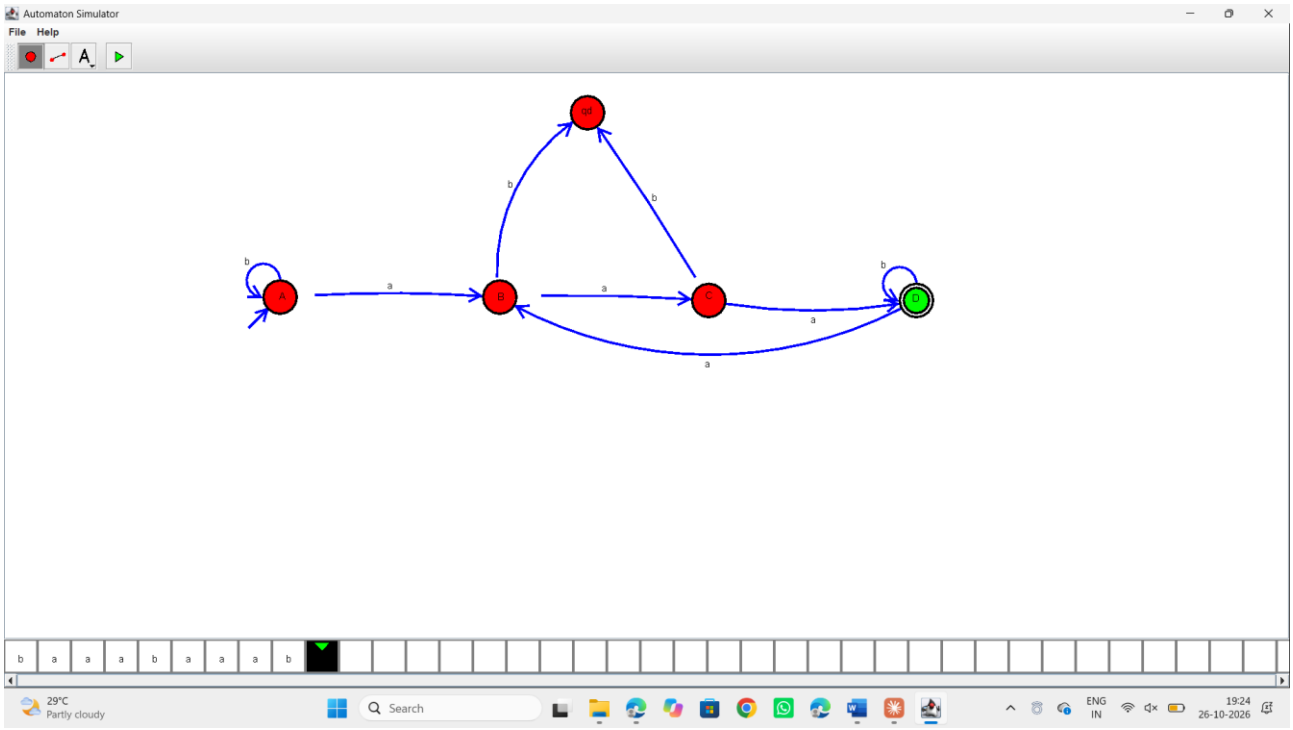

Automaton Simulator
File Help
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b c a a a

15. Draw a Deterministic Finite Automata for the language accepting strings ending with ’01’ over input alphabets ∑ = {0, 1}.

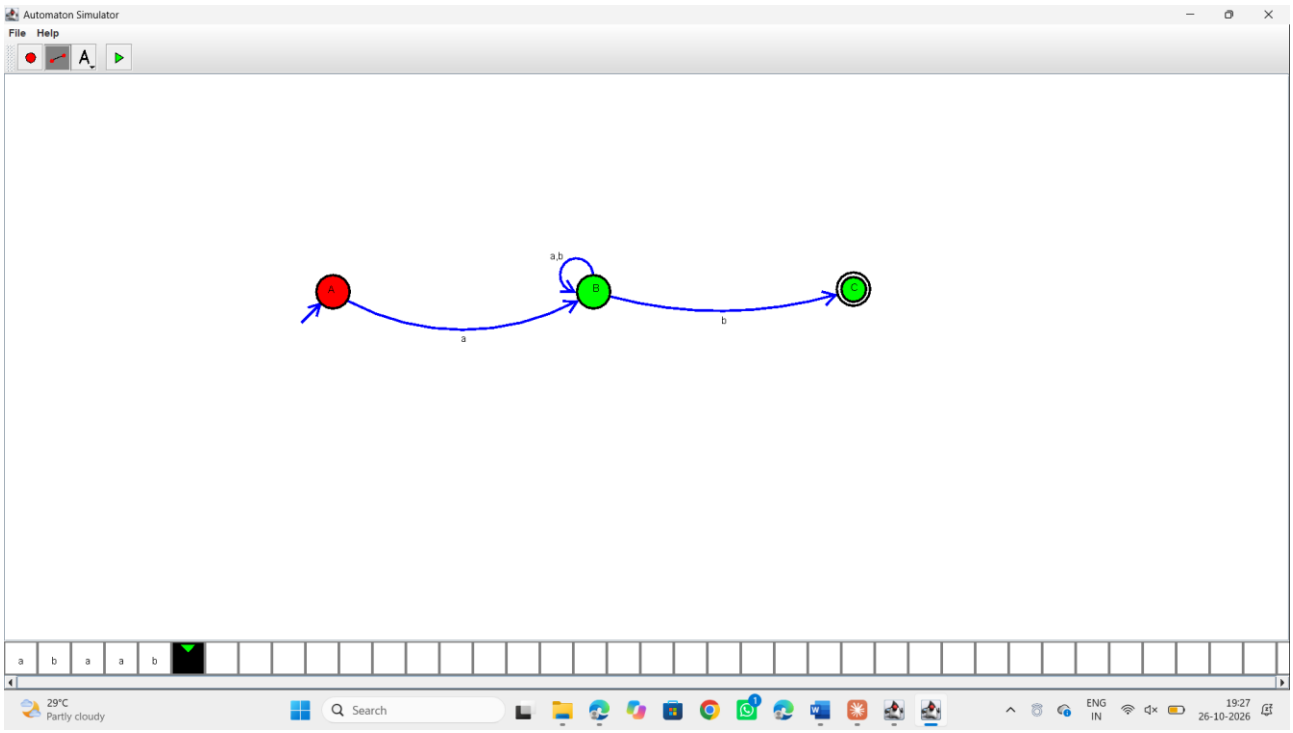
Here a=0 , b=1



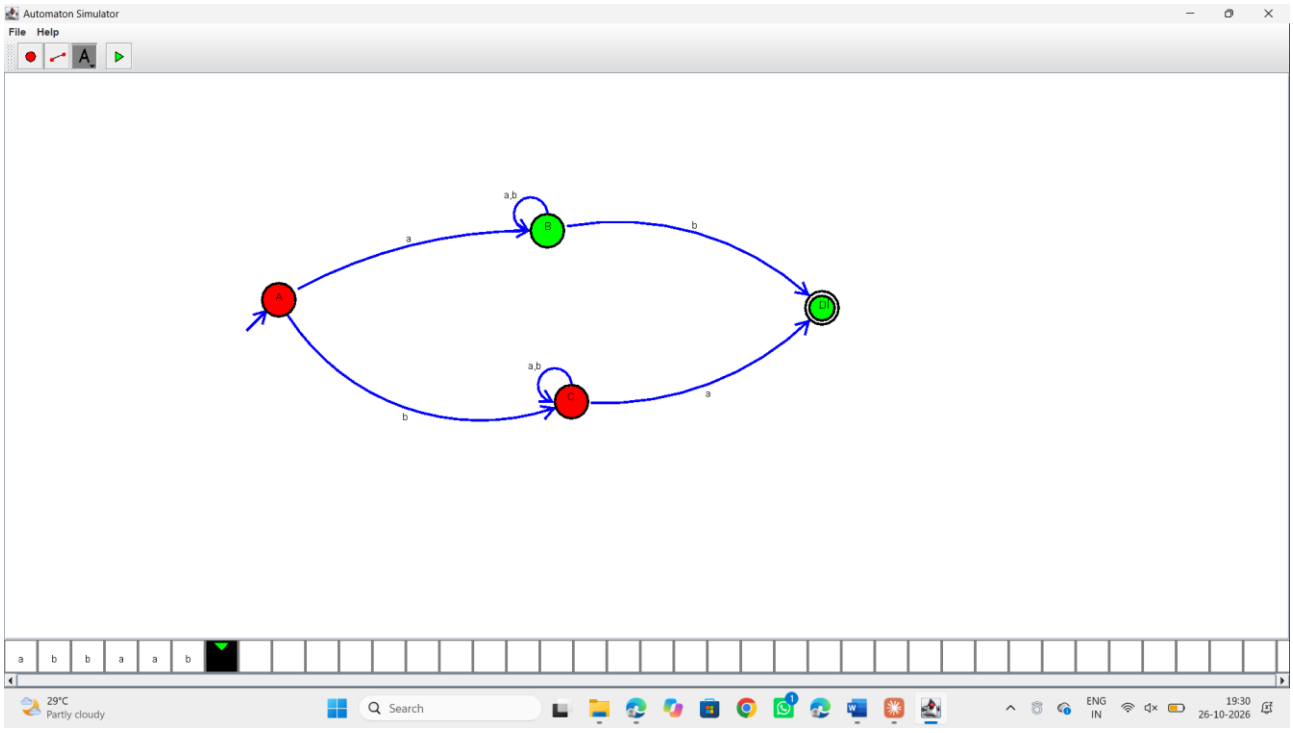
16. Design Deterministic Finite Automata using simulator to accept strings in which a’s always appear tripled over input {a,b}



17. Design Non Deterministic Finite Automata using simulator to accept the string the start with a and end with b over set {a,b} and check W= abaab is accepted or not.

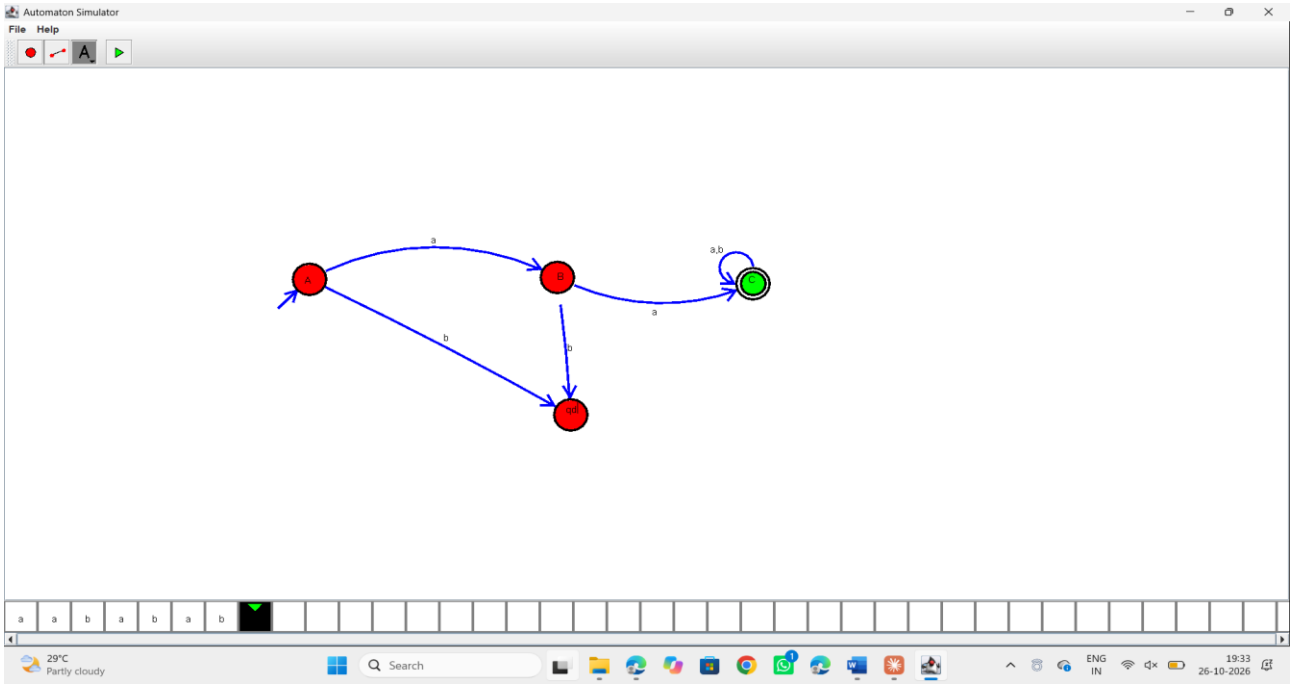


18. Design Non Deterministic Finite Automata using simulator to accept the string that start and end with different symbols over the input {a,b}.



19. Draw a Deterministic Finite Automata that accepts a language L over input alphabets ∑ = {0, 1} such that L is the set of all strings starting with ’00’.

Here a=0 , b=1



20. Design DFA using simulator to accept the string the end with abc over set {a,b,c) W= abbaababc

